Weather Radar App: How It Works

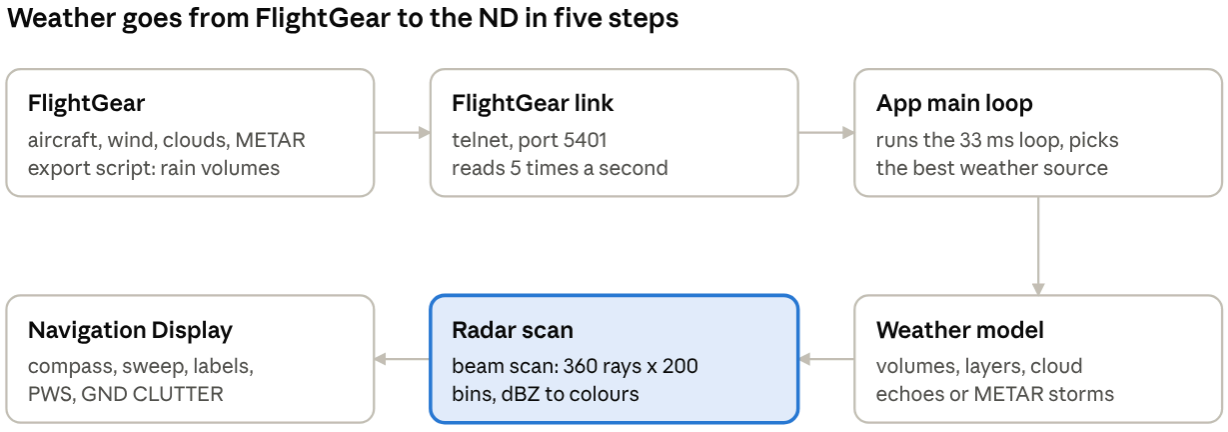
Overview

The app is an airborne weather radar that runs beside the FlightGear flight simulator and shows FlightGear's weather on a Navigation Display (ND). It is a Python desktop program (numpy for the radar maths, PySide6/Qt for the window).

It does four jobs:

1. **Read** the aircraft and the weather from FlightGear over a telnet link (port 5401), 5 times a second.
2. **Build a weather model** the radar can look into: FlightGear's rain/snow volumes, echoes on FlightGear's clouds, or storms made from the METAR.
3. **Scan** that weather with a simulated radar beam, the way a real antenna does: tilt, beam width, earth curve, ground, freezing level, rain attenuation.
4. **Draw** the result on the ND in the standard radar colours, with the sweep, range rings, compass, labels and windshear alerts.

It can also run on its own (Simulated mode) with built-in storm scenarios, and send a METAR to FlightGear so FlightGear's weather matches the app.



data flow · FlightGear to the ND

The radar scan (highlighted) is the heart of the app; everything before it prepares the weather, everything after it draws the picture.

Weather radar and ND basics

An airborne weather radar sees **water drops and snow, not clouds**. The antenna in the nose sends short radio pulses ahead, sweeps left and right, and measures how much energy comes back from each distance. More and bigger drops send back more energy.

Reflectivity (dBZ) is that returned energy on a log scale. Light rain is about 20 dBZ, a thunderstorm core 50+ dBZ. The ND paints it in fixed colours:

Colour	dBZ	Meaning
Black	below 20	nothing worth showing
Green	20 to 30	light rain
Yellow	30 to 40	moderate rain
Red	40 to 50	heavy rain, avoid
Magenta	50+	extreme rain; in WX+T mode also turbulence
Cyan	weak	snow (in this app) or weak ground in MAP mode
White		rain and snow mixed (in this app)

The beam has a height. It is about 3.5° wide and goes out in a straight line while the earth curves away below it. So at 80 NM it is thousands of feet thick and may pass over or under a storm. The pilot aims it with **tilt** (up or down, in degrees).

The ground also echoes. If the beam hits the ground, it shows as clutter that looks like weather. Pilots tilt up until the ground just disappears at the edge of the screen. Mountains also shadow what is behind them.

Heavy rain blocks the beam (attenuation). A strong cell in front can hide weather behind it, so a black area behind red is not always safe.

Freezing level. Above it, rain is ice crystals, which echo weakly (about 7 dB less). Just below it, melting snow echoes stronger (the "bright band").

The ND (Navigation Display) shows the radar picture over a compass arc, track up. Range rings mark distance; the selected range (10 to 320 NM) is the outer arc. The left side shows the radar mode, the tilt, and the gain setting in cyan.

Radar modes:

- **OFF / STBY**: no beam, nothing shown.
- **TEST**: a fixed pattern of colour bands to check the display.
- **WX**: weather.

- **WX+T**: weather plus turbulence (magenta) within 40 NM, from Doppler.
- **MAP**: aimed down at the ground for coastlines and mountains.

Gain: CAL is the calibrated setting; manual gain makes echoes stronger or weaker (shown as VAR).

PWS (Predictive Windshear): below 2,300 ft it looks 5 NM ahead for microbursts and warns with an icon, a WINDSHEAR message, and a voice.

Where the weather comes from

FlightGear has two weather systems, and the radar reads each one differently. Basic Weather has no storm positions, so the app builds a picture that matches it. Detailed Weather has real rain areas, so the radar shows exactly those.

A real weather radar sees only rain and snow, never dry cloud. The app follows the same rule wherever it can.

Part 1: Basic Weather

Basic Weather sets the same weather everywhere around the aircraft. It comes either from a METAR (live, a scenario, or typed in) or from Manual Configuration (cloud layers and sliders). FlightGear gives no position for any storm or rain area.

1a. Basic Weather from a METAR: label **FG METAR** (amber)

The app reads the METAR text and builds storms that match it:

In the METAR	What the radar shows
TS, CB or TCU	Scattered thunderstorms: about 7 cells, 3 to 8 NM wide, 15 to 140 NM ahead, red cores with turbulence
RA or SN (no TS)	One wide rain area about 60 NM ahead with 2 stronger cells inside it
No rain (e.g. FEW, SCT only)	Nothing
+ (heavy) or - (light)	Stronger or weaker storms
FEW, SCT, BKN, OVC	How many cells (more cover, more cells)
SN, or RA and SN	Snow shows cyan; mixed shows white

The type and strength are right. The positions are made up, because a METAR does not say where each storm is. That is why the label is amber. The storms drift with FlightGear's wind.

1b. Basic Weather, Manual Configuration: label **FG LAYERS** (cyan)

The app reads the cloud layers and the rain and snow sliders:

FlightGear setting	What the radar shows
Rain or snow slider above 0	Rain (or snow) patches; stronger slider, stronger echo
Cloud cover (few 20%, scattered 45%, broken 75%, overcast 100%)	The patches cover that share of the screen
Highest raining layer top	Rain reaches from the ground up to it; the beam above it sees nothing
Turbulence table by altitude	Turbulence at the height the beam looks at (WX+T)
Temperature table	The freezing level, where rain turns weaker (ice)
Rain slider at 0	Nothing, even with clouds: dry cloud does not show

Patches have soft edges and are heavier inside, and they drift with the wind.

Part 2: Detailed Weather

Detailed Weather builds real clouds in tiles about 40 x 40 km around the aircraft. For every cloud that rains, it also makes a **rain volume**: an area (circle, ellipse or rectangle) with a base and top altitude and a rain, snow and turbulence value. Inside FlightGear, this volume is what makes it rain on the windscreen.

2a. Volumes: label **FG VOLUMES n** (cyan), the normal case

The export script inside FlightGear (wxradar_export.nas) copies all volumes every 2 seconds. It loads by itself each time FlightGear starts. The radar then shows each volume exactly where it is, as high as it reaches, as strong as FlightGear set it.

FlightGear rain value	Colour on the radar
under 0.07	nothing

FlightGear rain value	Colour on the radar
0.07 to 0.32	green
0.32 to 0.57	yellow
0.57 to 0.82	red
over 0.82	magenta

Turbulence above 0.5 inside rain shows magenta in WX+T mode, within 40 NM, so every FlightGear thunderstorm (turbulence 0.6, 0.8 or 1.0) shows it.

What each weather type gives (from FlightGear's own code):

Weather type	Rain volumes per tile	On the radar
Thunderstorms	1 or 2 storms: circles 3 to 5 km radius, ground to 20,000 ft, rain 0.8 to 1.0, turbulence 0.6 to 1.0; plus a few light showers (rain 0.1 to 0.3)	1 or 2 red or magenta storms, a few faint green showers
Low pressure	1 tile in 5: 2 ellipses about 11 NM long, rain 0.3 outside and 0.5 inside. 1 in 5: a 15 NM square, rain 0.1. 3 in 5: no rain	Green and yellow areas in about 1 tile of 5; faint green or nothing in the rest
High pressure	No rain	Nothing

That is why there can be 60 clouds but only 2 to 4 volumes: most clouds do not rain, and the radar correctly shows only the ones that do.

Each volume has one rain value for the whole area, so a storm shows as one even colour rather than green edges around a red core.

2b. While Detailed Weather is building: label **METAR TEMP** (amber)

For a few seconds after you press Apply, or after a reset, FlightGear has no clouds or volumes yet. The radar shows storms from the scenario's METAR (as in 1a) until the volumes arrive, then switches to them.

2c. Cloud echoes, the backup: label **CLOUD ECHO n** (amber)

Used only if the export script is not running. The app puts a 2 to 3.5 NM echo on every cloud, with a strength guessed from the weather type: high pressure 0, low pressure 0.7, cold front 1.1, thunderstorms 1.2. Every cloud gets an echo, rain or not, so the picture looks full but is not FlightGear's real rain.

Tile Selection Mode

This FlightGear setting decides what weather is built as you fly. The radar just shows the result.

Mode	What FlightGear builds
single tile	One tile only; no weather once you fly out of it
repeat tile	The same type again and again (the Thunderstorm scenario's default)
realistic weather	The next tile can be a different type, e.g. low pressure then cold front (the pressure scenarios' default)
METAR	Tiles built from the METAR

How the radar scan is computed

Once the weather is known, the radar looks into it the way a real antenna does: point by point, at the height the beam really is.

The scan grid

The screen is a half circle ahead of the aircraft, 90° left to 90° right.

Term	Meaning	Value
Ray	One line out from the aircraft at one angle	360 rays, one every 0.5°
Bin (beam point)	One distance along a ray	200 bins, from the aircraft to the selected range
Beam sample	The beam is 3.5° thick, so each point is checked at several heights	7 heights, the centre counting most
Sweep	The line that moves across the screen	45° a second, about 4 s from one side to the other

One full picture is 360 x 200 = 72,000 points, each checked at 7 heights. Only the rays the sweep just passed are worked out again, about 30 times a second, so the picture repaints behind the sweep like a real radar.

Steps for each point

1. **Where is it?** The point's position from the ray angle, the aircraft's track and the distance.
2. **How high is the beam there?** Aircraft altitude + distance x tilt + the drop of the earth's curve (the ground falls away below a level beam). Worked out for all 7 heights.
3. **Is the beam in the ground?** The height is compared with FlightGear's real terrain there (or level ground where none is known). Parts of the beam below the ground give ground echo, not weather.
4. **Is there rain or snow here, at this height?** The weather model from the section above answers:
    - **Volumes**: is the point inside a volume, between its base and top? Then that volume's rain, snow and turbulence.
    - **Cloud layers** (Manual Configuration): is the point in a rain patch, below the top of the raining layer?
    - **METAR storms or cloud echoes**: how strong are the cells around the point? These have no height from FlightGear, so the app gives them one: stronger cells reach higher, up to about 42,000 ft.
5. **Freezing level**: above it, rain is ice and echoes about 7 dB weaker; just below it, melting snow echoes stronger (the bright band).
6. **Terrain shadow**: once the beam has hit the ground or a mountain, nothing behind it is seen.
7. **Attenuation**: heavy rain (over 30 dBZ) weakens everything behind it along the ray.
8. **Gain and smoothing**: add the gain setting and a little noise, then soften the picture so edges look natural.
9. **Colour**: dBZ to colour: green from 20, yellow from 30, red from 40, magenta from 50; snow cyan, mixed white. In WX+T, turbulence within 40 NM turns magenta. In MAP mode the ground shows in cyan, yellow and magenta.
10. **Ground clutter**: points where ground echo is stronger than weather are remembered, so the screen can mark them GND CLUTTER.

Why height matters

The radar shows rain only where the beam actually passes through it. Two examples:

- **Low pressure rain** reaches only from the ground to the cloud base, a few thousand feet. At 25,000 ft the beam is far above it near the aircraft and meets it only far away. Tilt down, or fly lower, to see it.
- **Thunderstorms** reach 20,000 ft, so they show at almost any altitude and tilt.

Auto tilt sets the tilt so the bottom of the beam touches the ground just past the selected range, in 0.25° steps. The screen then shows weather without ground clutter. On the ground it aims the beam to just graze the runway.